

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to Consider
Streamlining Interconnection of Distributed
Energy Resources and Improvements to
Rule 21.

Rulemaking 17-07-007
(Filed July 13, 2017)

**REPLY COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON
VEHICLE TO GRID ALTERNATING CURRENT INTERCONNECTION SUBGROUP
REPORT**

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Dated: January 13, 2020

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In accordance with Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”)¹ hereby submits its reply to comments on the Vehicle-to-Grid Alternating Current (“V2G AC”) Interconnection Technical Sub-Group Final Report (“Final Report”) submitted in Rulemakings 17-07-007 and R. 18-12-006 on January 6, 2020. Pursuant to *Joint Administrative Law Judges’ Ruling Establishing Subgroup and Schedule to Develop Proposal on Mobile Inverter Technical Requirements for Rule 21 and Noticing Workshop*, issued by Administrative Law Judges Kelly A. Hymes, Sasha Goldberg, and Patrick Doherty on August 23, 2019, VGIC timely files these reply comments on January 13, 2020 in R. 17-07-007 and R. 18-12-006. VGIC also previously filed a motion for party status in these proceedings on January 6, 2020.

I. INTRODUCTION

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., Enel X North America, Inc., Fiat Chrysler Automobiles, Ford Motor Company, General Motors Company, and Nuvve Corporation. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all of the individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>).

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles (“EVs”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and discharging is recognized and compensated in support of achieving a more reliable, affordable, and efficient electric grid. VGIC appreciates the opportunity to participate in this proceeding to inform and shape the development of a pathway to V2G AC interconnection.

II. REPLY COMMENTS

Reply to the California Energy Storage Alliance’s (CESA) Comments:

VGIC generally supports CESA’s comments and provides the following additional response:

- VGIC is supportive of CESA’s concept of an “initial phase” of V2G interconnections that could be pursued under UL 9741 pathway. This pathway was also described in VGIC’s initial comments submitted on January 6, 2020. However, VGIC cautions that pursuing such a pathway in the near-term should not foreclose other interconnection pathways from being pursued over the long-term, including pathways that do not require any third-party testing procedures and could instead utilize OEM internal testing procedures.
- VGIC is supportive of CESA’s recommendation to reconvene the subgroup once updates to the necessary standards (e.g. SAE J3072, UL 9741) have been completed. VGIC believes these updates will coincide with the suggested July 2020 timeframe as discussed in the December 17, 2019 workshop. However, VGIC is open to some flexibility in the exact date of reconvening the subgroup if there are unforeseen delays in completing updates to the

necessary standards. VGIC also agrees with CESA that reconvening the subgroup should not be contingent on the availability of EV test results.

- VGIC also supports CESA’s recommendation for the Commission to consider the jurisdictional questions regarding internal certification through the OEM testing procedures. In its initial comments, VGIC outlines several potential pathways for how this might work to satisfy CPUC and IOU needs. For example, one option may be to rely upon governmental entities such as the National Highway Traffic Safety Administration (“NHTSA”). To that end, VGIC offers to work with stakeholders over next several months to provide more detail on how NHTSA’s procedures could work in support of a potential pathway for internal testing.

Reply to Southern California Edison’s (SCE) Comments:

VGIC offers the following comments in response to SCE’s initial comments:

- VGIC generally supports SCE’s conceptual timeline outlined on page 2 of its comments. However, VGIC believes the timeline could be accelerated and recommends that the end point for the “initial phase” of V2G AC Implementation could occur as early as the mid-2021 timeframe (rather than early 2022).
- VGIC opposes SCE’s requirement for third-party testing. VGIC recognizes that third-party testing could be an option if pursued through UL 9741 as described in the Final Report, however this should be seen only as one possible pathway rather than an absolute requirement. VGIC agrees with SCE that eliminating safety risk is paramount and that third-party certification has been a successful way to achieve this for other grid-connected devices in the past. However, this approach is not feasible for automobiles, which are subject to a

more complex and sophisticated manufacturing environment². Furthermore, VGIC believes the requirement to use third-party testing is a restrictive approach that undervalues and misunderstands the rigor already involved in automotive grade testing. For example, section 7-D-1 of the Final Report details the following aspects of how internal testing currently operates:

- “OEMs will validate their product to ensure they can achieve the relevant certification by meeting all FMVSS, EPA, and other standards and requirements. This self-certification step is typically a continual, 18-month process spanning from the design freeze until the beginning of production.”³
- “The magnitude of effort, internal support resources, and test facilities to perform validation and revalidation is significant, as this process occurs as frequently as each month during the product development cycle.”⁴

The ability to develop and fully test products internally is also critical to advancing products over a short development cycle, and to ensure the security of competitive information during product development. Given these factors, VGIC continues to support OEM internal testing as a potential pathway for validating the safety and security of interconnected V2G equipment. This could include accreditation of OEMs as a Nationally Recognized Testing Laboratory (“NRTL”) as suggested by VGIC in its initial comments and by the joint comments of SDG&E and PG&E.

² VGIC agrees with CESA’s statement on page 9 of their initial comments that “mobile inverters are wholly integrated into the software, firmware, and hardware of the vehicle, which may play many different functions beyond those for smart inverters.”

³ Final Report, p 45

⁴ Final Report p 46

Reply to the Joint Comments of San Diego Gas and Electric (SDG&E) and Pacific Gas and Electric's (PG&E)

VGIC offers the following comments in response to SDG&E and PG&E's joint initial comments:

- VGIC opposes SDG&E and PG&E's recommendation for requiring EV test results for UL 1741 prior to reconvening the subgroup. Such "pre-qualification" testing would require the OEMs to incur significant expenditures to prepare application-specific test equipment and develop new software, firmware and hardware to meet test specifications, some of which may only apply to off-board/stationary equipment that is not applicable to PEVs. Since this testing would not be directly linked to an interconnection pathway, such an approach would only be practical if there were an external source of funding available to all OEMs to support this activity.
- VGIC supports SDG&E and PG&E's concept of a potential interconnection pathway whereby OEMs become accredited as a NRTL for EV testing. This is very similar to several of the pathways presented in VGIC's initial comments, including number 2 (see page 5 of VGIC initial comments) and number 5 (see page 6 of VGIC initial comments).

Respectfully submitted,



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